Yussif Sophyani Banaamwini

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Professional Profile

I am a computer science-oriented researcher and educator with experience in machine learning, computer vision, and human-computer interaction. I possess strong technical expertise in deep learning and intelligent system development, complemented by practical teaching, mentorship, and publication experience. I am a committed collaborator with a strong work ethic, eager to add value to research and academic institutions.

Research & Teaching Interests

- Machine/Deep Learning
- Human Activity Recognition
- Human-Computer Interaction (HCI)
- Natural Language Processing (NLP)
- Computer Vision (medical imaging, remote sensing)
- Game Design & Game Development

Education

Ph.D. Computer Science and Technology, University of Electronic Science and Technology of China — Sep. 2020 – Dec. 2024

MEng. Computer Science and Technology, University of Electronic Science and Technology of China — Sep. 2018 – Jun. 2020

BSc. Mathematics with Economics, University of Cape Coast — Aug. 2013 – May. 2017

Professional Experience

Postdoctoral Fellow – School of Computer Science and Engineering (School of Cyber Security), University of Electronic Science and Technology of China, China — Jan. 2025 – present

RESPONSIBILITIES

- Advancing digital media technology research through the development of action recognition and human-computer interaction systems for industrial applications.
- Engage with faculty members and research partners to identify and develop research projects that align with the center's objectives.
- Assist in instructing three (3) courses on machine learning and game development, including:
    1) ***Human-Computer Interaction***
    2) ***Game Design***
    3) ***Game Development***
- Assist in supervising and mentoring graduate students (***3 PhD*** and ***1 Master***), offering expert guidance and fostering research excellence across their respective projects.
- Instructing and participating in academic sessions and seminars to facilitate knowledge exchange and interdisciplinary collaboration.

Post-Graduate Teaching/Research Assistant – School of Computer Science and Engineering (School of Cyber Security), University of Electronic Science and Technology of China, China — Sep. 2020 – Nov. 2024

RESPONSIBILITIES

- Conducted in-depth research to improve human activity recognition by introducing novel frameworks and deep learning models, publishing in top conferences and journals.
- Supported the full research publication cycle alongside academic staff and laboratory researchers (***4 PhD students***), contributing to experimental execution, data analysis, and dissemination of findings through peer-reviewed journals and conferences.
- Tracked and evaluated students' academic progress against milestones, initiated corrective measures, and mentored undergraduates to enhance research and writing skills.
- Assisted in teaching, evaluating, and grading the postgraduate course ***Multimedia Technology and Applications***, ensuring fair, rigorous assessment in line with established academic standards.
- Collaborated with external colleagues to design and execute research, analyze data, and prepare findings for peer-reviewed journals.

Graduate Research Assistant – School of Computer Science and Engineering (School of Cyber Security), University of Electronic Science and Technology of China, China — Sep. 2018 – May 2020

RESPONSIBILITIES

- Spearheaded innovative research initiatives in computer vision, with a focus on designing and implementing human activities recognition and human-computer interaction systems tailored for real-world deployment.
- Cultivated strategic collaborations with laboratory staff to conceptualize and drive research projects that are aligned with the center's mission and long-term objectives.
- Collaborated with external research partners to identify and develop research projects, including conducting experiments, data analysis, and preparing manuscripts for publications.
- Participated in building Virtual-Reality (VR) projects, including incorporating multiplayer mechanisms in virtual scenes.

Assistant Secretary – Dean's Office, Faculty of Social Sciences, University of Cape Coast, Ghana — Sep. 2017 – Aug. 2018

RESPONSIBILITIES

- Served as Recorder and Acting Secretary to the Faculty Appointments and Promotions Committee, drafting agendas, recording minutes, communicating decisions to relevant officers, and reporting on action progress.
- Supervised the Faculty Officer's Office, managing the Faculty Officer's itinerary, maintaining confidential files, and handling official correspondence to ensure efficient administrative operations.
- Administered the Faculty Student Internship Programme, coordinating placements, corresponding with students and faculty, and ensuring the successful execution of internship schedules.
- Oversaw student admissions screening by evaluating and selecting prospective students in accordance with approved cut-off limits.
- Provided comprehensive administrative support to the Dean and Faculty Officer, executing additional assigned duties and ensuring the overall operational efficiency of the Dean's Office.

Publications

#	Title	Author Order	Journal/Conference Name	Year
1	EMNet: Efficient Multimodal Network for Action Recognition	1	Information Fusion **[JCR Q1]**	2026
2	Self-relational graph convolution network for skeleton-based action recognition	1	Proceedings of the 31st ACM International Conference on Multimedia **[CCF A, ICORE A*]**	2023
3	EGTCN: An efficient graph and temporal convolution network for sensor-based human activity recognition in federated learning	1	IEEE Sensors Journal **[JCR Q1]**	2024
4	Bridging the gap: Predictive contracts in blockchain - achieving recalibration for industrial networks	2	Journal of Industrial Information Integration **[JCR Q1]**	2024
5	Flood or Non-Flooded: A Comparative Study of State-of-the-Art Models for Flood Image Classification Using the FloodNet Dataset with Uncertainty Offset Analysis	2	Water **[JCR Q2]**	2023
6	MCDF: Multimodal information fusion and causal analysis for election misinformation detection	3	Information Fusion **[JCR Q1]**	2026
7	Tb-mmrd: transformer-based multi-modal election rumor detection with agreement-aware gating and semantic fusion	3	Multimedia Systems **[JCR Q2]**	2025
8	Animal species detection and classification framework based on modified multi-scale attention mechanism and feature pyramid network	3	Scientific African **[JCR Q1]**	2022
9	Pre-Trained Transformer-Based Models for Text Classification Using Low-Resourced Ewe Language	3	Systems **[JCR 1]**	2024
10	LiSENCE: A Hybrid Ligand and Sequence Encoder Network for Predicting CYP450 Inhibitors in Safe Multidrug Administration	3	Mathematics **[JCR 1]**	2025
11	Gaze-it: Intelligent multimodal AR cross-reality control system	3	IEEE International Symposium on Mixed and Augmented Reality Adjunct **[CCF B, ICORE A*]**	2023
12	DCAGTN: Dual-snapshot causal-attention graph transformer with feedback mechanism for real-time rumor detection	4	Knowledge-Based Systems **[JCR Q1]**	2026
13	Cov-Fed: Federated learning-based framework for COVID-19 diagnosis using chest X-ray scans	4	Engineering Applications of Artificial Intelligence **[JCR Q1]**	2024
14	E-MuLA: An Ensemble Multi-Localized Attention Feature Extraction Network for Viral Protein Subcellular Localization	4	Information **[JCR Q2]**	2024
15	MEN: leveraging explainable multimodal encoding network for precision prediction of CYP450 inhibitors	4	Scientific Reports **[JCR Q1]**	2025
16	CMNet: Cross-Modal Coarse-to-Fine Network for Point Cloud Completion Based on Patches	5	IEEE Transactions on Circuits and Systems for Video Technology (IEEE TCSVT) **[JCR Q1]**	2025

#	Title	Author Order	Journal/Conference Name	Year
17	Learning-centered multi-modal federated learning complimented with filter rotation for clinical image diagnosis	5	Engineering Applications of Artificial Intelligence **[JCR Q1]**	2025
18	Classification from Sky: A Robust Remote Sensing Time Series Image Classification Using Spatial Encoder and Multi-Fast Channel Attention	5	Journal of King Saud University - Computer and Information Sciences **[JCR Q1]**	2022
19	Automated vision-based structural health inspection and assessment for post-construction civil infrastructure	7	Automation in Construction **[JCR Q1]**	2023

For more information: ORCID Google Scholar

Academic Activities

Peer Reviews

- IEEE Transactions on Circuits and Systems for Video Technology (IEEE TCSVT)
- Information Fusion
- Alexandria Engineering Journal (AEJ)
- Scientific Reports
- Engineering Applications of Artificial Intelligence (EAAI)
- ACM International Conference on Multimedia (ACM MM)

Seminars/Workshops/Knowledge Dissemination

- Conducts seminars for graduate researchers, offering guidance on research methodologies, scientific writing, and effective research practices during postdoctoral and PhD study.
- Delivered a multi-day Keras (TensorFlow) deep learning workshop series at UESTC, providing hands-on training to graduate researchers.
- Informally mentors and guides research colleagues on the practical use of machine learning frameworks, supporting the implementation of research projects through hands-on technical assistance.

Awards & Honors

- PhD degree Scholarship, University of Electronic Science and Technology of China, 2020-2024.
- Academic Achievement Award (second prize), University of Electronic Science and Technology of China, 2018-2019.
- Excellent Performance Award (second prize), University of Electronic Science and Technology of China, 2018-2019.
- Master's degree Scholarship, University of Electronic Science and Technology of China, 2018-2020.
- Winner of "UESTC Graduation Sports-2019" football game.

Technical Projects

AI Glasses Project – Automated system for car industry assembly. Feb. 2026 – present

RESPONSIBILITIES

- Co-led development of an automated worker supervision system, setting the technical roadmap and integrating AI and XR components.
- Developed deep learning models to automatically detect and analyze worker activities during assembly, enabling intelligent, real-time supervision of task execution.
- Cooperating with laboratory researchers to integrate smart glasses (XR)-based interfaces into the supervision pipeline, facilitating real-time visual feedback and hands-free worker performance evaluation.

AI English Learning Application – AI-powered English language learning application for children with multilingual support. May 2024 – present

RESPONSIBILITIES

- Developing an AI-powered English language learning application for children by leveraging NLP and large language model (LLM) technologies to enable intelligent speech-to-text and text-to-speech functionalities, supporting interactive and adaptive learning experiences.
- Integrating multilingual instructional support into the application using NLP/LLM-based models, enabling children from diverse linguistic backgrounds to receive guided English learning instructions in their native languages.
- Developing an interactive, child-friendly front-end interface (UI/UX) for the AI-powered application using Unity Engine, ensuring an engaging and age-appropriate user experience.
- Collaborates with backend developers to seamlessly integrate intelligent features such as adaptive content delivery and interactive feedback into the front-end environment.
- Optimizes application performance, visual responsiveness, and cross-platform compatibility to ensure smooth and consistent user experiences across target devices before and after user testing.

"Pilolo" Game – 3D mobile game deployed on the Play Store. Jun. 2017 – Jul. 2018

RESPONSIBILITIES

- Engineered core gameplay mechanics, game logic, and interactive systems through end-to-end programming, ensuring optimized performance and responsive controls tailored for the Android platform.
- Created all 3D assets, character models, and environmental textures, maintaining visual consistency, artistic quality, and performance efficiency throughout the game environment.
- Designed and implemented an intuitive UI/UX system, including menus, HUD elements, and in-game interfaces, to deliver a seamless and engaging player experience on mobile devices.
- Rigged and created characters, UI, and object animations within the game scene, ensuring fluid motion, visual cohesion, and an immersive interactive experience across all animated elements.
- Integrated a virtual reality (VR) mode to deliver an immersive extended reality experience within the game environment.
- Managed the full Android deployment pipeline, including build configuration, performance optimization, compliance with Google Play Store policies, and successful public release of the application.

Skills

Programming & Frameworks

- **C#** scripting for Unity Engine game and application development, including gameplay mechanics, UI systems, and interactive 2D/3D environments.
- **Python** for machine and deep learning research utilizing PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, Numpy, OpenCV, NLTK, etc.
- **C++** for low-level programming tasks in minor computational applications and system-level development.
- **HTML & CSS** for designing and structuring basic front-end interfaces and web-based components in minor application development projects.

Development Tools

- **Unity Engine** for the design and development of interactive 2D/3D games, VR, and real-time applications.
- **Blender** for 3D modeling and animation with applied experience in creating optimized game-ready assets and environments for real-time rendering pipelines.
- **Adobe Photoshop** for digital image editing, UI asset creation, and visual design, supporting the production of high-quality graphics for games and multimedia applications.
- **Visual Studio Code** (VS Code) is the primary code editor for writing, debugging, and managing Python and C# codebases across machine learning research and Unity Engine development projects.
- **Microsoft Office Suite** for document preparation, data organization, and professional correspondence.
- **Azure DevOps** for project management, version control, and collaborative software development workflows.

Languages

- **English** – Fluent, Official
- **Dagaare** – Native
- **Akan** – Native
- **Mandarin Chinese** – Basic

Interests

- Football
- Reading
- Mentoring
- Sightseeing
- Game and Application Development